

For **tourism and hotel management in Singapore**, digital transformation is driven by increasing competition for visitors, changing traveller expectations, and technologies such as AI, data analytics, mobile applications and contactless services.

Singapore is a particularly good case because it welcomed **16.9 million international visitors in 2025**, with tourism receipts of **S\$32.8 billion**. At the same time, STB is actively encouraging tourism businesses to adopt AI, data and other technologies.

1. Market Drivers — “Why does tourism/hospitality need to change?”

Market Driver	Singapore Example	Impact
Increasing global competition	Singapore competes with destinations such as Bangkok, Bali, Kuala Lumpur, Tokyo and Hong Kong.	Hotels and tourism businesses need differentiated experiences and better service.
Growing visitor numbers	Singapore received 16.9 million visitors in 2025.	Businesses need scalable ways to manage more visitors.
Changing travel trends	Travellers increasingly seek wellness, cultural, experiential and sustainable tourism.	Hotels develop new experiences beyond traditional accommodation.
Labour and productivity pressures	Hotels need to provide high-quality service while managing workforce constraints.	Encourages automation, AI and smart hotel systems.
Sustainability pressure	Hotels are expected to reduce energy, water,	Drives smart energy management and

	waste and carbon consumption.	sustainable hotel operations.
Competition from digital travel platforms	Customers can compare hotels, prices and reviews through online travel platforms.	Hotels need strong digital marketing and customer engagement.
Need for new revenue streams	Hotels increasingly offer wellness, dining, events and experiences.	Encourages digital platforms and cross-industry partnerships.

STB's Hotel Industry Transformation Map specifically identifies **digital-enabled experiences, sustainability, productivity and innovation** as important areas for the industry's development.

Example

A traditional hotel may focus primarily on:

Room + Restaurant + Customer Service

A transformed hotel may offer:

Room + Mobile App + Digital Check-in + Smart Room + Personalised Offers + Wellness + Experiences + Data Analytics

2. Technology Drivers — “What makes transformation possible?”

Technology Driver	Tourism/Hotel Example	Benefit
Artificial Intelligence (AI)	AI-powered travel assistants and hotel chatbots	Provides faster customer assistance

Generative AI	AI creates personalised travel recommendations and itineraries	Improves trip planning
Data analytics	Analysing visitor spending and booking behaviour	Enables targeted marketing
Mobile apps	Hotel apps for booking, check-in, room services and payments	Provides convenient self-service
Contactless check-in	Digital identity/facial recognition-based check-in	Reduces waiting time
IoT / smart rooms	Sensors control lighting, temperature and room equipment	Improves guest experience and energy efficiency
Robotics	Service robots and automated hotel operations	Reduces repetitive work
Cloud computing	Cloud-based property management systems	Connects hotel operations
AR/VR	Virtual tours and AR experiences	Allows visitors to explore destinations digitally
CRM systems	Tracking customer preferences and interactions	Enables personalised marketing

Singapore already provides a strong real-world example. STB reports that **28 tourism companies adopted technology solutions in 2025**, while AI was increasingly used to improve operations and reduce workforce workload. For example, Conrad Centennial Singapore's AI-powered scheduling system reduced scheduling time from two hours to 15 minutes.

STB has also highlighted contactless check-in using its **E-Visitor Authentication (EVA)** system, integrated with hotel self-check-in solutions.

3. Customer Drivers — “What do travellers expect?”

Customer Driver	Traveller Expectation	Digital Response
Convenience	“I want to book and manage everything from my phone.”	Mobile apps
Speed	“I don’t want to wait at reception.”	Mobile/contactless check-in
Personalisation	“Recommend activities that suit me.”	AI and customer analytics
24/7 assistance	“I want help whenever I need it.”	AI chatbots and virtual assistants
Seamless experience	“I don’t want to repeatedly provide my information.”	Integrated CRM and hotel systems
Real-time information	“Tell me about events, transport and attractions now.”	Mobile apps and digital platforms

Unique experiences	"I want more than just a hotel room."	Digital experiences, wellness and experiential tourism
Sustainability	"I prefer environmentally responsible travel."	Smart energy systems and digital sustainability initiatives
Flexible services	"I want to choose how I interact with the hotel."	Self-service + human service options
Digital payment	"I want a quick cashless payment experience."	Mobile and contactless payments

STB's current strategy specifically highlights using **data collaboration, traveller profiles and GenAI** to better understand and respond to changing traveller behaviour.

Putting the Three Drivers Together

Consider a **Singapore hotel**.

MARKET DRIVER

There is increasing competition for international travellers.



The hotel needs to differentiate itself and improve productivity.

TECHNOLOGY DRIVER

AI + mobile apps + IoT + data analytics + contactless technology



The hotel can provide more efficient and personalised services.

CUSTOMER DRIVER

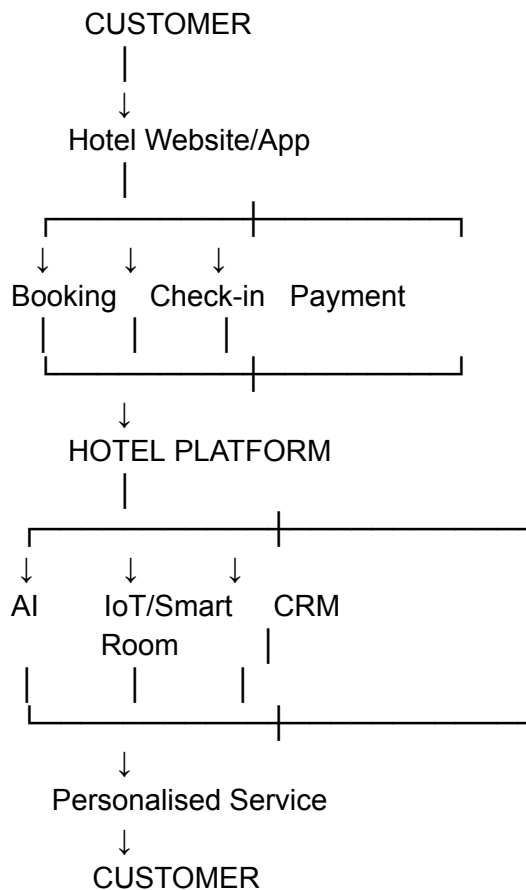
Travellers expect:

Fast + convenient + personalised + seamless experiences



DIGITAL TRANSFORMATION

Smart and connected hotel experience



Example: Smart Hotel in Singapore

Imagine a guest books a hotel room using a mobile application.

Before arrival

AI analyses preferences

“The guest previously selected a high-floor room.”



The hotel provides a personalised room recommendation.

Arrival

Digital check-in



Guest avoids a long reception queue.

Inside the hotel

IoT smart room

Guest uses the app to control:

- Air-conditioning
- Lighting
- Curtains
- Television

During the stay

AI concierge

Guest asks:

“What are the best attractions near my hotel?”

The system provides personalised recommendations.

After the stay

CRM

The hotel records relevant customer preferences and can provide personalised offers for future visits.

Singapore Tourism Example: AI

A particularly strong current example is the **Singapore Tourism Board’s partnership with OpenAI**. In July 2025, STB and OpenAI announced an MoU aimed at preparing Singapore’s tourism sector for an AI-driven future.

This demonstrates how AI can move tourism from:

Generic tourism information

to:

Personalised, intelligent travel assistance

For example:

Traditional:

“Here are 20 attractions in Singapore.”

AI-powered:

“You have two days, you’re travelling with young children, you prefer indoor activities and you’re staying near Orchard Road. Here’s a personalised itinerary.”

Easy Classroom Framework

MARKET

Why must tourism/hotel businesses change?

- Global competition
- Increasing visitor numbers
- Labour/productivity pressures
- Sustainability
- Changing travel trends
- Online travel platforms



TECHNOLOGY

What enables the change?

- AI
- Generative AI
- Mobile apps
- Data analytics
- IoT
- Robotics
- AR/VR
- Cloud
- Contactless technology



CUSTOMER

What do travellers want?

- Convenience
- Speed
- Personalisation
- 24/7 assistance
- Seamless experience
- Unique experiences
- Sustainability
- Digital payment



DIGITAL TRANSFORMATION

A connected, personalised and data-driven tourism and hospitality experience.